CHAPTER 7

MEASURES OF DISPERSION

CHAPTER ESSENCE AT A GLANCE:
Master absolute and relative measures of scatter: Range, Quartile Deviation, Mean Deviation, Standard Deviation, Variance, and Coefficient of Variation for comparing group consistency.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Meaning of Dispersion:** Dispersion measures the extent of scatter or variation of individual observations around a central value; absolute measures carry units, relative coefficients are dimensionless.

• **Range and Quartile Deviation:** Range is extreme difference (Max - Min); Quartile Deviation (semi-interquartile range) is based on middle 50 percent of data [QD = (Q3 - Q1) divided by 2] and resists outliers.

• **Mean Deviation:** Arithmetic average of absolute deviations from mean or median; ignores algebraic plus and minus signs, limiting its mathematical tractability.

• **Standard Deviation and Coefficient of Variation:** Standard Deviation (positive square root of variance) is the gold-standard measure; Coefficient of Variation [CV = (SD divided by Mean) * 100] compares consistency across series.

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$\text{Standard Deviation} \iff \sigma = \sqrt{\frac{\sum f_i (x_i - \bar{x})^2}{N}} = \sqrt{\frac{\sum f_i x_i^2}{N} - \bar{x}^2}$$

Root Mean Square Deviation Formulation

$$\text{Coefficient of Variation} \iff \text{CV} = \frac{\sigma}{\bar{x}} \times 100$$

Relative Dimensionless Measure of Consistency

$$\text{Quartile Deviation} \iff \text{QD} = \frac{Q_3 - Q_1}{2} \quad \text{and} \quad \text{Coefficient} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Semi-Interquartile Positional Scatter Measure

$$\text{Change of Scale Rule} \iff y = a + bx \to \sigma_y = |b|\, \sigma_x \quad [\text{Independent of Origin } a]$$

Linear Transformation Standard Deviation Invariant

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Standard deviation is independent of change of origin (adding or subtracting a constant leaves SD unchanged) but is dependent on change of scale (multiplying by b multiplies SD by |b|).

• **Alert:** When comparing two series, the series with the LOWER Coefficient of Variation is more consistent, stable, and uniform.

• **Alert:** Variance is the square of standard deviation; never report variance with standard deviation units (variance is squared units).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Absolute Measures and Relative Measures of dispersion with one example of each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Absolute Measures:** Expressed in the same physical units as the original observations (e.g. Standard Deviation in kg or cm). Cannot compare series with different units or widely different averages. **[1 Mark]**

• **Relative Measures:** Pure, dimensionless ratios or percentages independent of measurement units (e.g. Coefficient of Variation). Ideal for comparing dispersion between different datasets. **[1 Mark]**

2. Calculate the Standard Deviation and Variance of the following observations: 4, 8, 10, 12, 16. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Arithmetic Mean):** Sum of x = 4 + 8 + 10 + 12 + 16 = 50. N = 5. Mean = 50 divided by 5 = 10. **[1 Mark]**

• **Step 2 (Deviations from Mean):** (x - Mean) = -6, -2, 0, +2, +6. **[1 Mark]**

• **Step 3 (Squared Deviations):** $(x - Mean)^2$ = 36, 4, 0, 4, 36. Sum of squared deviations = 36 + 4 + 0 + 4 + 36 = 80. **[1 Mark]**

• **Step 4 (Variance and SD):** Variance = 80 divided by 5 = 16. Standard Deviation = square root of 16 = 4. **[1 Mark]**

3. Two cricket batsmen A and B scored the following average runs and standard deviations in a series: Batsman A: Mean = 50, SD = 10; Batsman B: Mean = 60, SD = 15. Determine: (a) Who is the higher run scorer? (b) Who is more consistent? [Score: 4 Marks]

Answer & Key Valuation Steps:

• **(a) Higher Run Scorer:** Batsman B has a higher mean score (60 > 50), so Batsman B is the higher run getter on average. **[1 Mark]**

• **(b) Consistency Evaluation (Coefficient of Variation):**
• CV for A = (SD divided by Mean) * 100 = (10 divided by 50) * 100 = 20 percent.
• CV for B = (SD divided by Mean) * 100 = (15 divided by 60) * 100 = 25 percent. **[2 Marks]**

• **Conclusion:** Since Batsman A has a lower Coefficient of Variation (20 percent < 25 percent), Batsman A is more consistent in batting performance. **[1 Mark]**

4. In a distribution, the lower quartile Q1 is 30 and the upper quartile Q3 is 50. Calculate: (a) Quartile Deviation, (b) Coefficient of Quartile Deviation. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **(a) Quartile Deviation:** QD = (Q3 - Q1) divided by 2 = (50 - 30) divided by 2 = 20 divided by 2 = 10. **[1.5 Marks]**

• **(b) Coefficient of QD:** Coefficient = (Q3 - Q1) divided by (Q3 + Q1) = (50 - 30) divided by (50 + 30) = 20 divided by 80 = 0.25. **[1.5 Marks]**

5. If the Standard Deviation of variable X is 6, find the Standard Deviation of variable Y defined as: Y = 3X + 5. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Transformation Rule:** Standard Deviation is completely independent of change of origin (the constant +5 has no effect) and depends directly on change of scale: SD(Y) = |b| * SD(X). **[1 Mark]**

• **Calculation:** SD(Y) = 3 * 6 = 18. **[1 Mark]**

6. State one major drawback of Mean Deviation and explain why Standard Deviation is preferred. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Major Drawback of Mean Deviation:** It artificially ignores algebraic negative signs by taking absolute values |x - Mean|, which is mathematically unsound and precludes further algebraic analysis. **[1 Mark]**

• **Why SD is Preferred:** Standard deviation squares the deviations to eliminate negative signs mathematically, making it capable of rigorous algebraic treatment and sampling distribution analysis. **[1 Mark]**